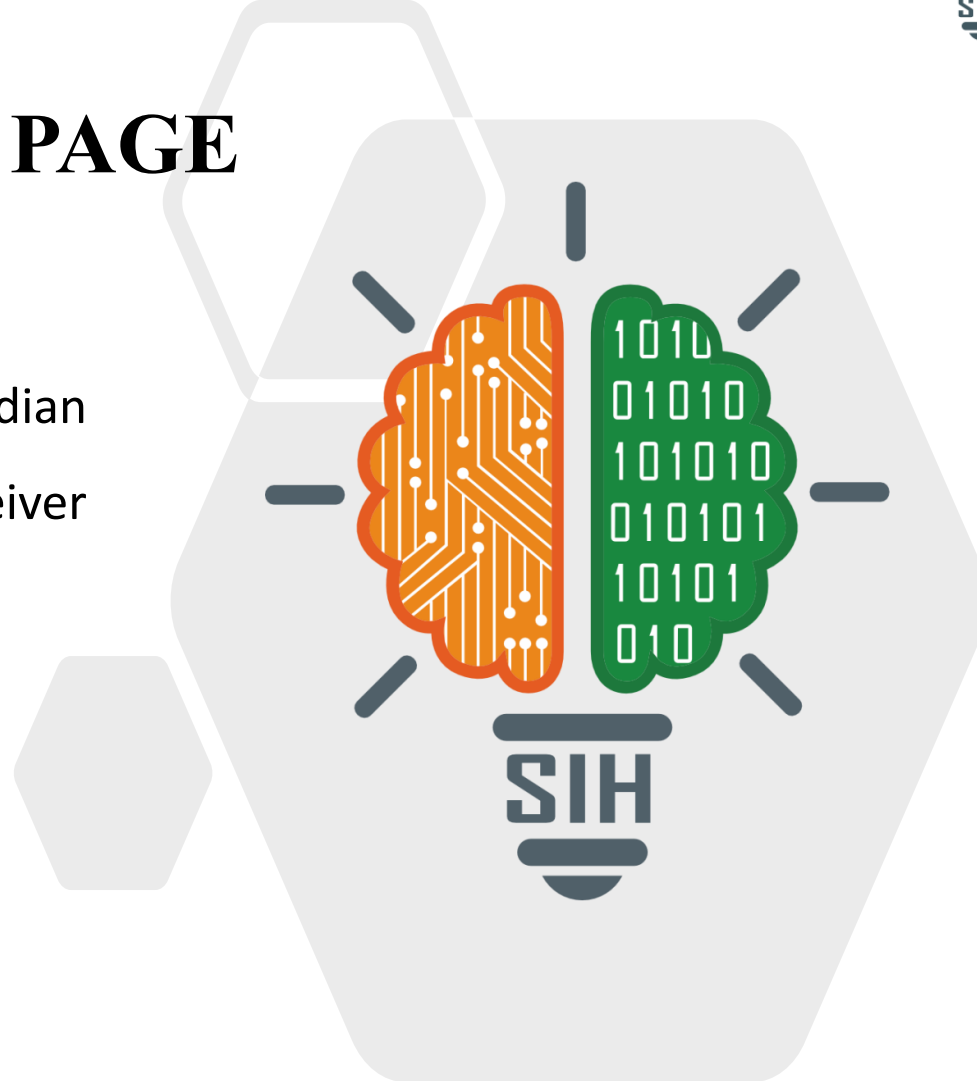


TITLE PAGE

- **Problem Statement ID :- 26173**
- **Problem Statement Title :- iTantra - Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links**
- **Theme :- Smart Automation**
- **PS Category :- Software**
- **Team ID :-**
- **Team Name :- Falcons**



Problem

- ❖ **Raw voice** is highly data-intensive for **low-bitrate** links, while **conventional** STT/TTS often **depends** on **internet** or cloud infrastructure.
- ❖ In **emergencies and remote areas**, this makes fast, inclusive, **multilingual** voice communication **difficult** when **connectivity is unavailable**.

Our Idea

- ❖ **iTantra converts speech into text locally**, transmits only compact data over **Wi-Fi/Bluetooth**, and reconstructs it as **speech** on the **receiving** device.
- ❖ It enables **low-latency, multilingual, offline** voice communication between **phones without** relying on **cellular or cloud infrastructure**.

Unique Value Proposition

- ❖ Transmits **speech as compact text packets instead of raw audio**, enabling voice communication over extremely low-bandwidth links.
- ❖ Provides **10+ language on-device STT/TTS** with no cloud dependency during operation, keeping communication private and functional offline.
- ❖ Enables **direct phone-to-phone communication through Wi-Fi Direct with Bluetooth fallback**, with near-instant STT-to-TTS relay for emergency use.

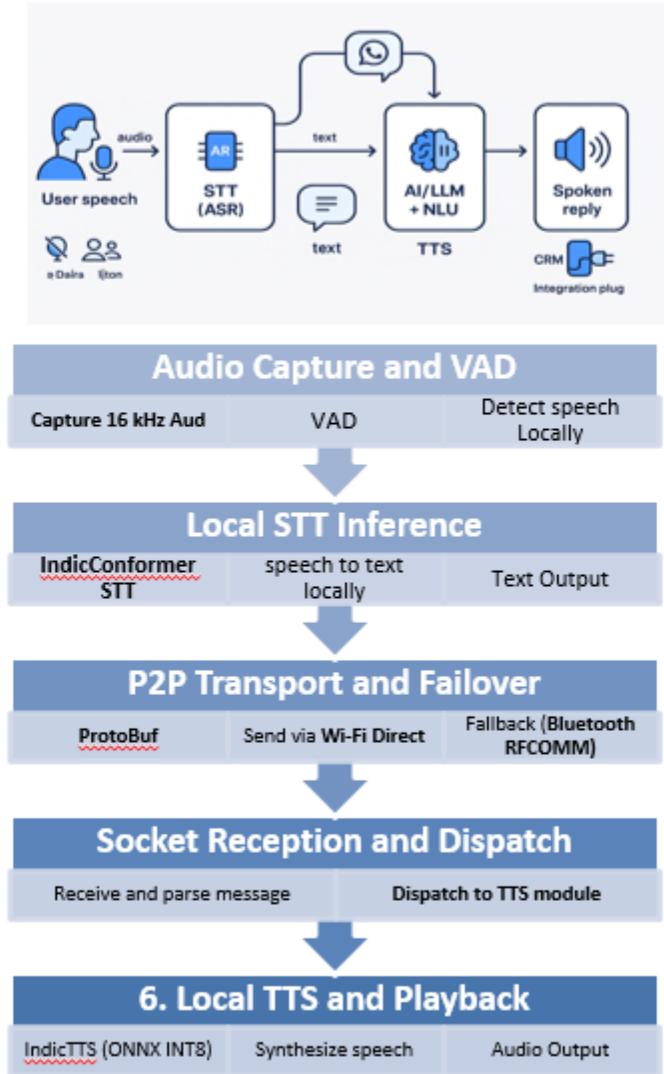


Proposed Solution

- ❖ Captures speech in short chunks and **automatically detects pauses** to finalize utterances.
- ❖ Performs **on-device STT and compact Protobuf packetization**, minimizing the data sent between devices.
- ❖ Uses **Wi-Fi Direct as the primary transport** and **Bluetooth RFCOMM as fallback** for infrastructure-independent communication.

TECHNICAL APPROACH

- ❖ **Hardware & Runtime:** Runs on low/md-range Android devices using mic, speaker, Wi-Fi and Bluetooth, with **INT8** models accelerated by **ONNX Runtime / NNAPI**.
- ❖ **Speech Processing:** Captures **16 kHz** mono PCM in **100 ms chunks**, uses **800 ms** pause-aware VAD, with **IndicConformer STT + VITS TTS** through **sherpa-onnx**.
- ❖ **Data & Communication:** Encodes speech text into compact **Protobuf v3** frames and transmits via **Wi-Fi Direct** / **WifiP2pManager**, with **Bluetooth RFCOMM** fallback.
- ❖ **Android Software Stack:** Built with **Kotlin + Jetpack Compose**, using **MVVM/Clean Architecture**, **Coroutines**, **Room**, **DataStore** and **Foreground Service**.



90% Fully Functioning Prototype Provided.
 Demo: <https://youtu.be/SBWvj8v4LYA>
 Our Codebase: <https://github.com/Precise-Goals/iTantra-Communication-Ecosystem>

Feasibility

- **Hardware & Runtime:** Runs locally on mobile chipsets using **ONNX Runtime Mobile and NNAPI**.
- **Storage & Memory:** **INT8** models require low storage and **controlled RAM usage**.
- **RF Connectivity:** Uses built-in Wi-Fi Direct and Bluetooth RFCOMM.
- **Open-Source Licensing:** Uses **Apache 2.0** and **MIT-licensed** technologies.

Challenges, Risks & Mitigation

- ❖ **Zero Infrastructure Cost:** **No cloud, internet and servers** subscriptions required.
- ❖ **Power Efficiency:** **VAD** reduces STT/TTS processing during silence.
- ❖ **Secure Offline Design:** Operates **without internet or cloud** dependency.
- ❖ **Inter-Agency Scalability:** **Protobuf** supports standardized **communication** between teams



Practical Implementation

- **Resilient Ad-Hoc Networking:** **Automatically pairs devices** via Wi-Fi Direct with a Bluetooth fallback, **ensuring priority SOS** messages are delivered **immediately** through high-volume alarm audio.
- **High-Speed Offline Edge AI:** A **Push-to-Talk interface** runs all speech processing entirely on-device, delivering translated, synthesized audio in under 2.5 seconds with zero internet dependency.



Benefits of the solution

01



Instant Deployment

Works without cellular, satellite, or Wi-Fi infrastructure.

02



Bandwidth Reduction

Reduces communication data requirements by over 98%.

03



Low Latency

Provides speech communication with under 2.5-second latency.

01



Multilingual Interoperability

Enables communication across different regional languages.

02



Critical Alert Delivery

Ensures priority SOS alerts receive immediate playback.

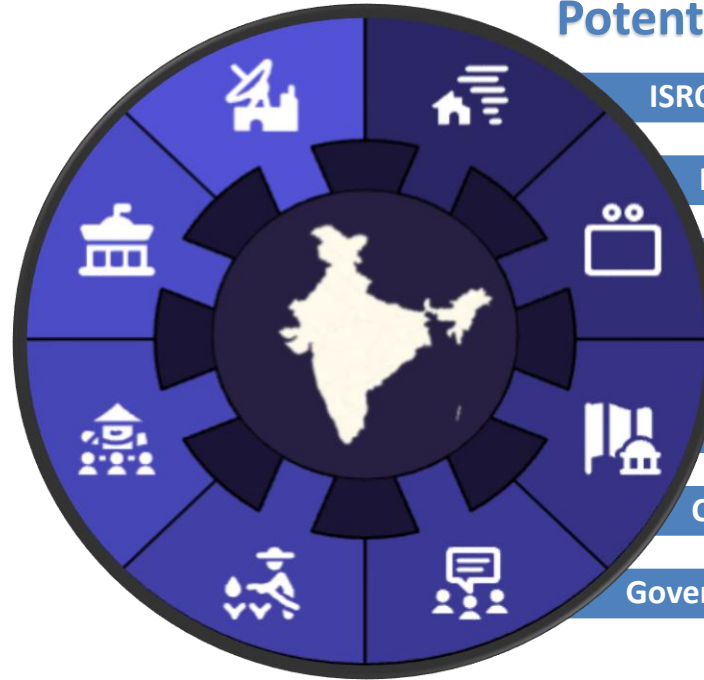
03



Reduced Cognitive Load

Converts distorted speech into clear synthesized audio.

Potential Impact of our Solution



ISRO: Shadow zone telemetry.

First Responders: Zero-infrastructure coordination.

Defense: Air-gapped operational security.

Nation: Reduces import dependency.

Task Forces: Eliminates language friction.

Communities: Stronger local trust.

Government: Simplified live reporting.

Future Scope

- ❖ **Deep-Space Telemetry & LoRa Expansion:** Compressing voice into tiny **50-byte** payloads positions **iTantra** perfectly for **ISRO's** satellite networks, while future **LoRa** hardware integration will extend our **secure, offline terrestrial** mesh beyond **10 kms**.
- ❖ **Accelerating India's Rural Economy: Transforming** everyday smartphones into **decentralized** communication hubs bypasses **vulnerable cellular grids**, ensuring **remote** farmers remain **permanently connected** to drive India's **national growth**.

Semantic Communication & Offline Edge AI:- *Associated Academic Study:* Real-Time **Speech-to-Text** on Edge: A Prototype System for **Ultra-Low** Latency and Local Processing - [MDPI Article](#)

Resilient Ad-Hoc Networking

- ❖ the Framework for **Satellite** Communication Network – [TRAI Paper](#)
- ❖ *Associated Academic Study:* IEEE **Device-to-Device** Wireless **Protocols** – [IEEE Published Research Paper](#)
- ❖ Low Bitrate High-Quality RVQGAN-based Discrete Speech Tokenizer – [IBM Research](#)
- ❖ Performance evaluation of low-bitrate voice using spread spectrum techniques for satellite-based emergency communication – [Science Direct Article](#)
- ❖ *Associated Academic Study:* An **Infrastructure-Less** Communication Platform for Android Smartphones Using **Wi-Fi Direct** – [Springer Natural Article](#)
- ❖ *Associated Academic Study:* Wi-Fi Direct in Android: Creating seamless **device-to-device** communication – [ArdaScience Article](#)
- ❖ Our Patent Research draft: iTantra Universal Communicator – [iTantra Article](#)

Project Ecosystem & Repositories

- ❖ Our Own Codebase: [Our Github Project Repository](#)
- ❖ Deployed Application Link: [Application Download URL](#) & [Application Release Page](#)
- ❖ Video Demonstration: [Prototype Youtube Video Link](#)
- ❖ Live Web Portal: [Landing Page for the Application](#)